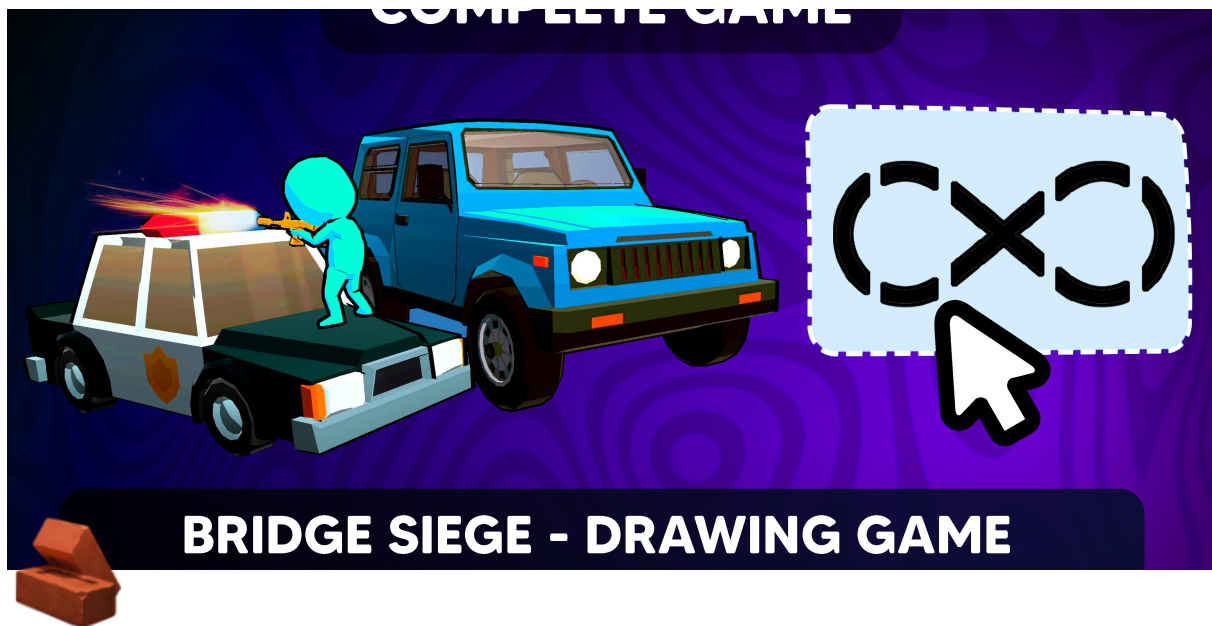




Bridge Siege

Our official stores

 Unity Asset Store

 Itch.io

 Fab

Recommended Unity Versions

Unity 2021.3 (LTS) /

Unity 2022.3 (LTS) /
Unity 6000.0.10f1

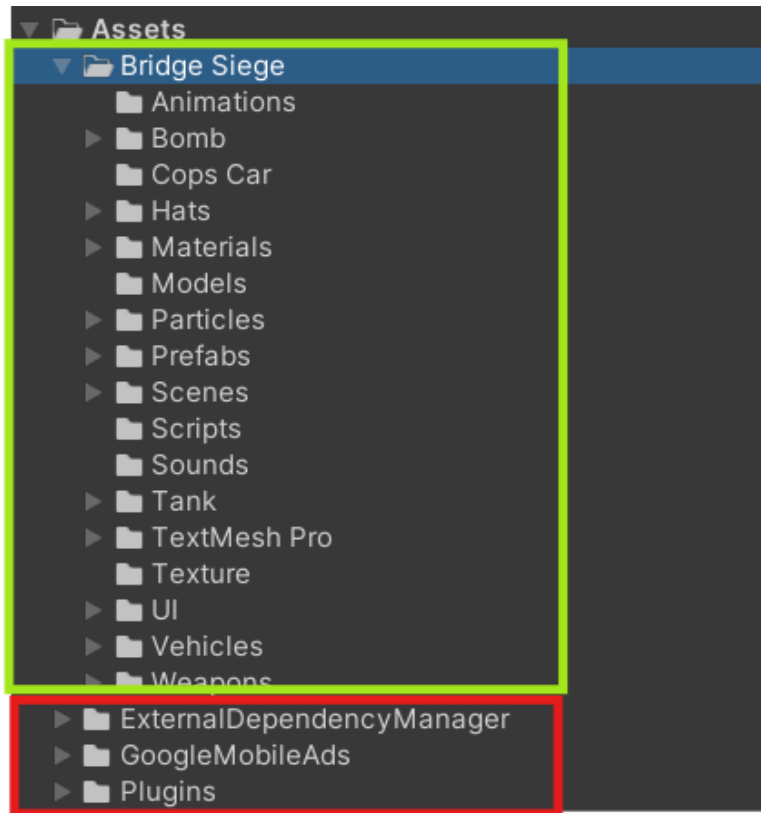
Project Template Import

- [Check here](#) > How to Import the Asset Pack Project

Prerequisites

- Import Google Mobile Ads SDK  > [Check here](#)

Project Structure



GAME TEMPLATE ASSETS

GOOGLE MOBILE ADS SDK

(Note: We Are using Unity's TMPPro for our Project)

We have everything inside the `Bridge Siege - Template` Folder

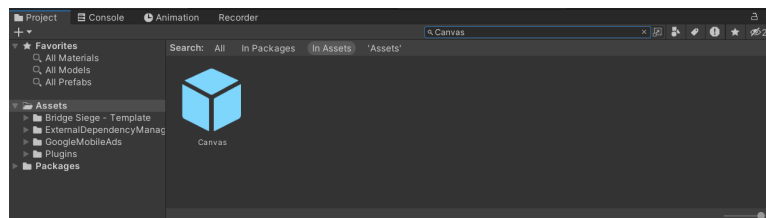
Components

Game Manager

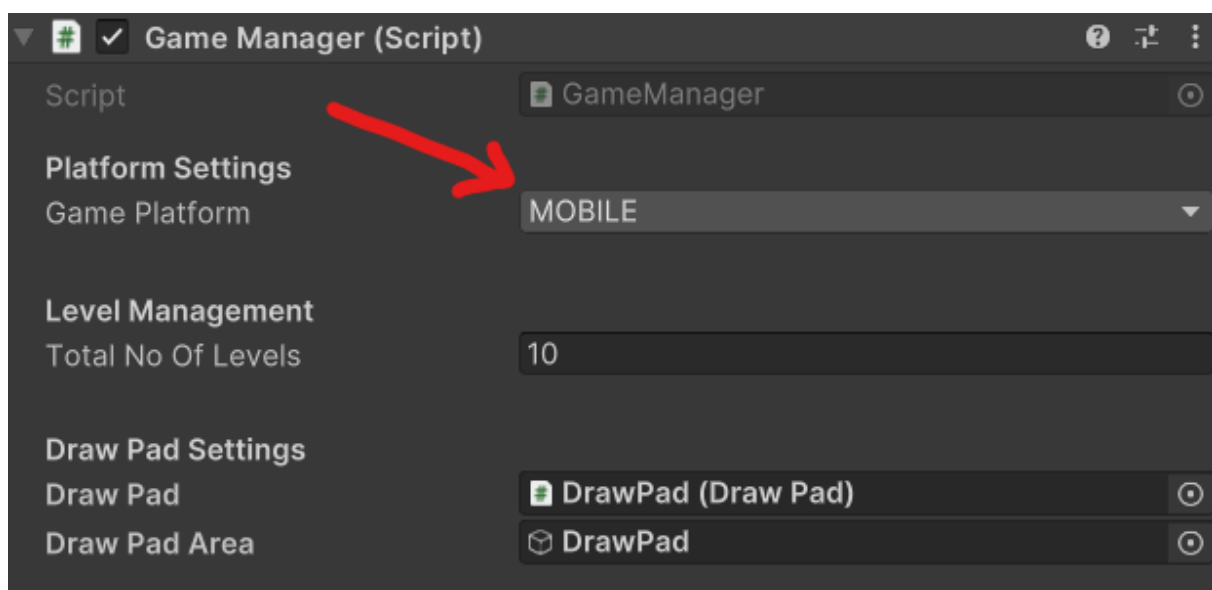
First Component is Game Manager, one of the most important components of our game, which is attached to CANVAS Prefab in every scene.

Game Manager is used to control the Level Progression, Level Loading, UI like

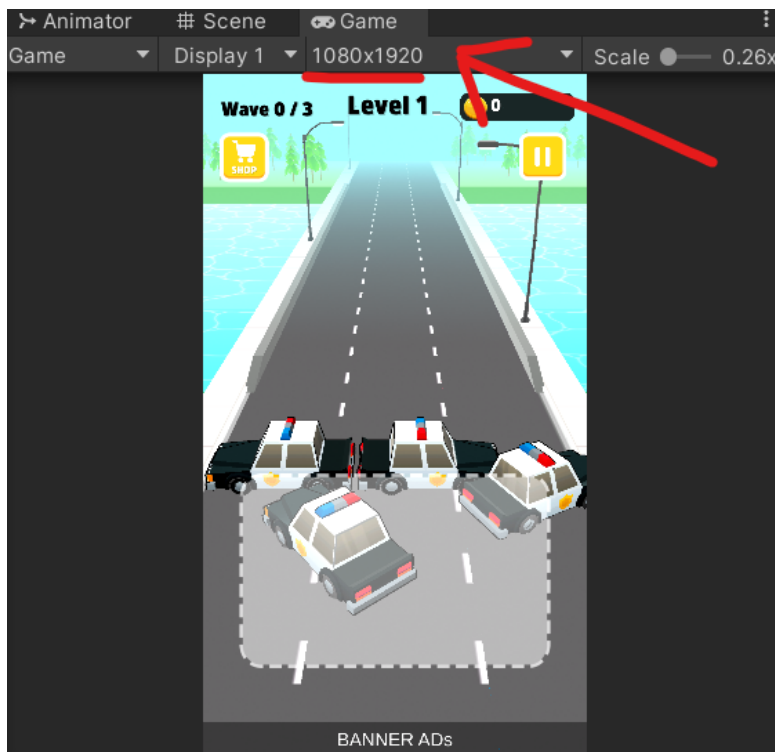
Win / Fail Screens, Rewards through Ads, Interstitial Ads and Pause / Resume Screen.



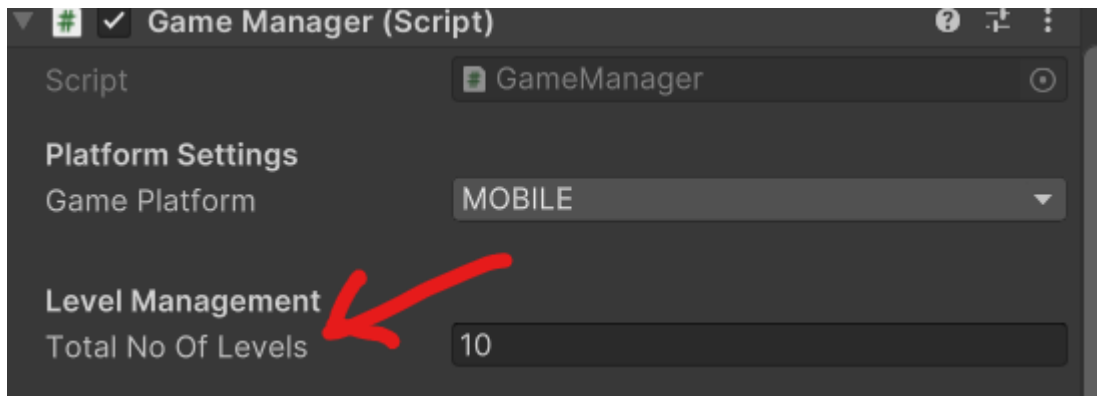
In the Project Panel, search for Canvas and open it up. Remember whatever changes you do in this Canvas Prefab will apply to all the Canvas Objects that are there in every scene.



Select your desired **platform** for which you want to build the game, and also change the **Game Window** resolution based on that. I want to build for Mobile so we will select **Mobile** and change **Game Window's Resolution** to **1080x1920**, but if you want to build for **PC and WebGL (Landscape)** select **PC** option respectively.



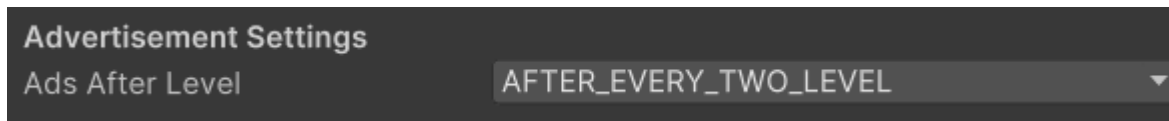
Now we have `Total No of Levels` variable, and you have to change it according to your needs.



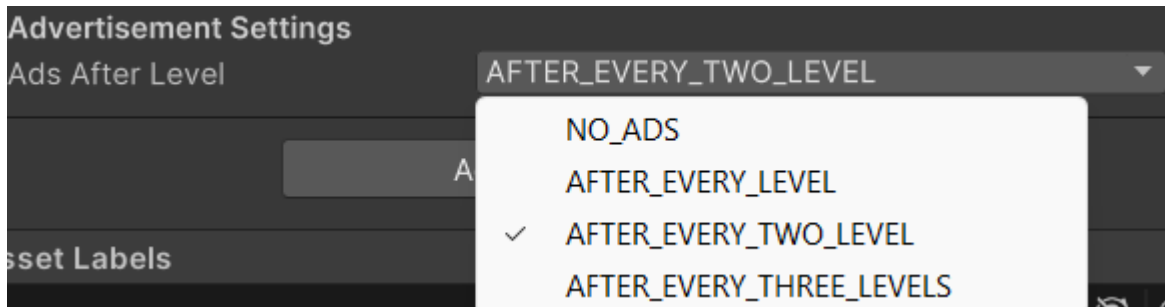
We have 10 Levels for our game, but you can create more, we're gonna show you how in this documentation.

You need to Tweak this variable after you have added new levels, also don't forget to add your new levels to Build Settings.

There are some other variables that store the references of majorly the UI Objects, another important thing that you should focus on is at the bottom.

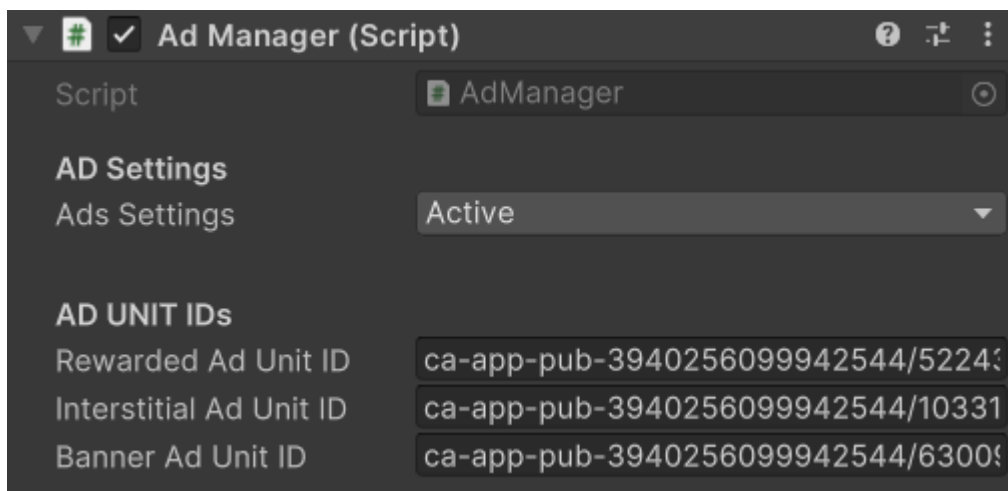


Here you can control the Frequency of Interstitial Ads



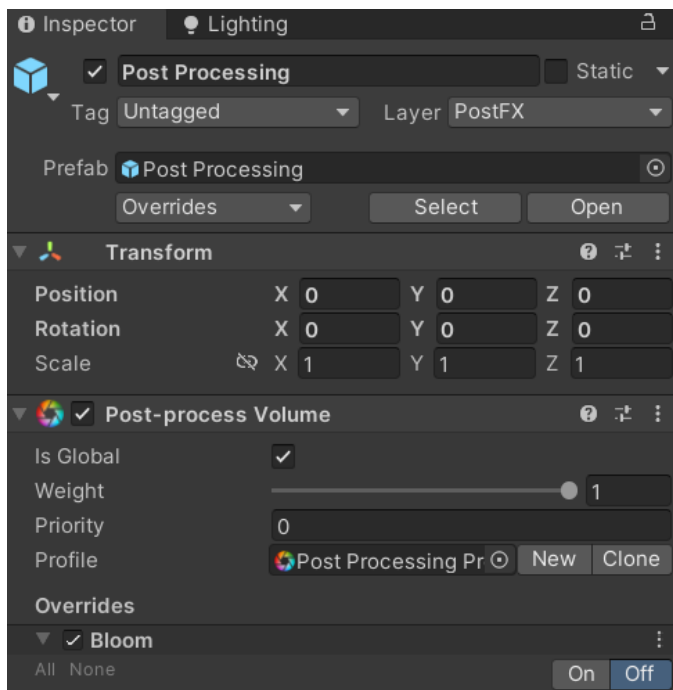
AdManager

Then we have another component, which is essential for showing ads to users, remember we have integrated the Google Mobile Ads SDK in our project.

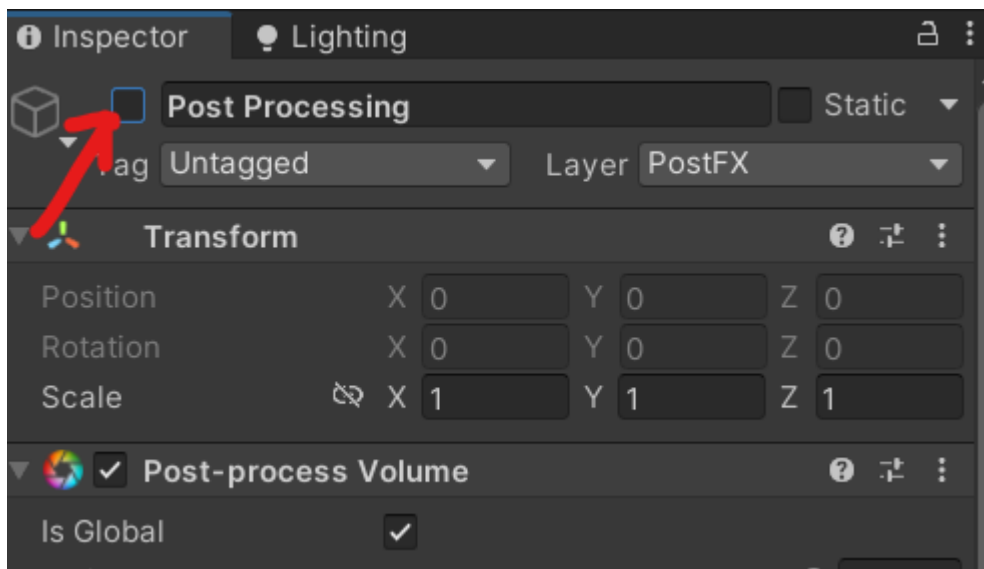


It's very self-explanatory actually so, you can set Ads Settings to Active or Inactive, also please insert your ad ids for rewarded video, interstitial, and banner ads in their respective slots.

Post Processing

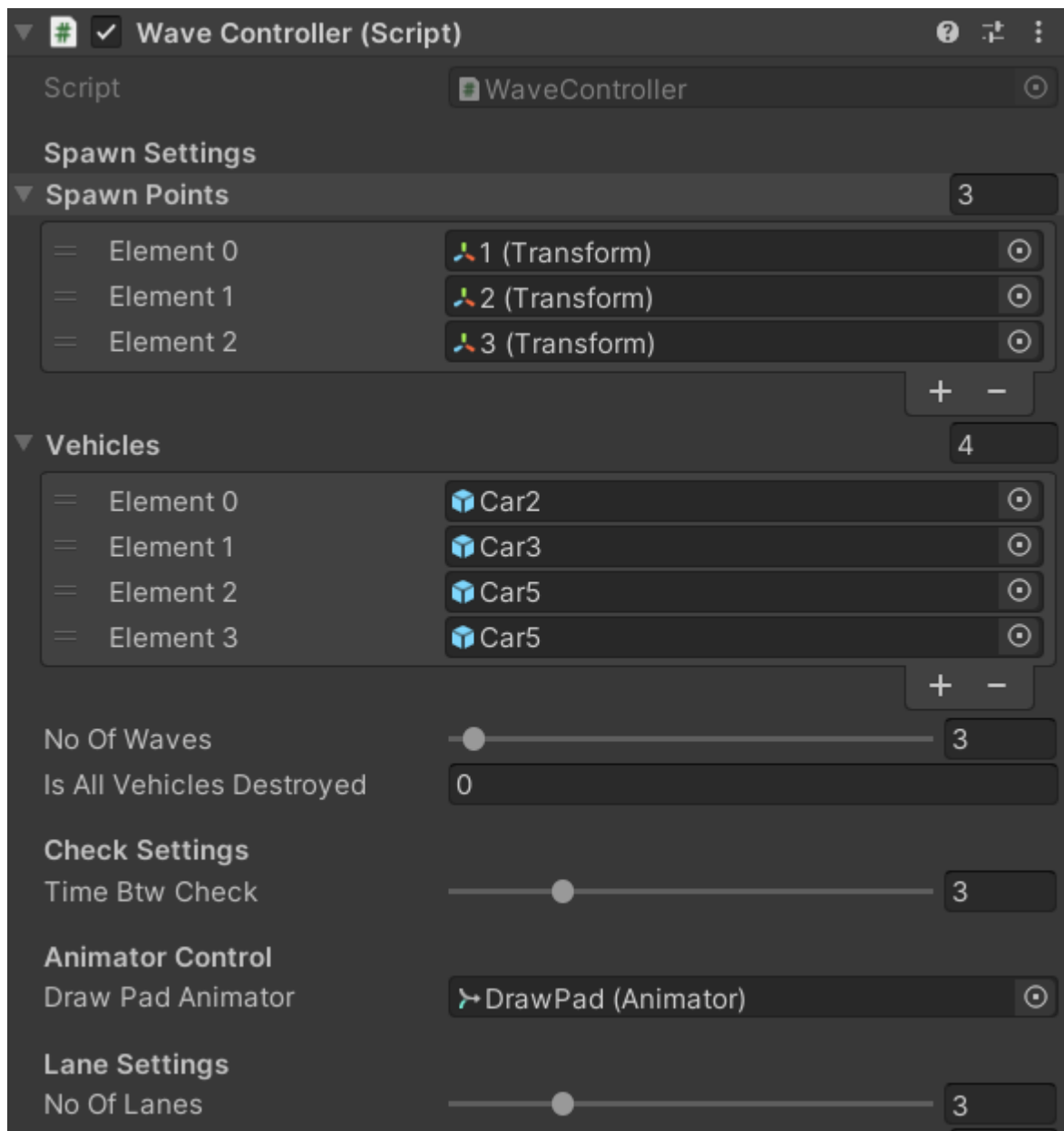


This Component is useful for enhancing the Visual Appeal of the game, if you want to build the game for **Mobile Audience** and you want extra **10-20 FPS**, Disable this Prefab Object.

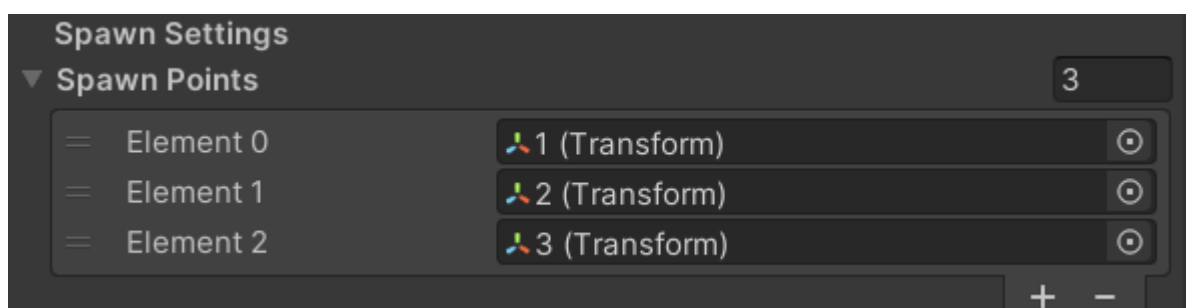


Wave Controller

This Component Controls the Waves of Vehicles that will arrive.

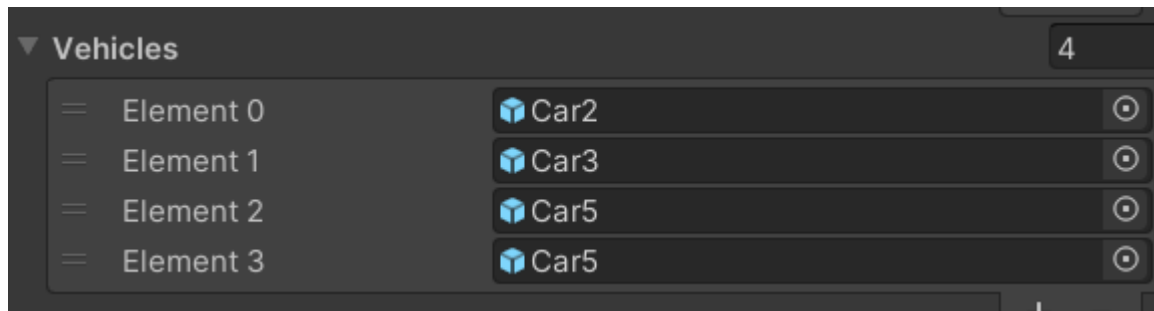


First, we have the Spawn Points, these are the points from where the new vehicles will be spawned, as we have a 3 lane system for this level (level 1) we have 3 Spawn Points.



Ideally, you should have Spawn Points equal to your lanes.

Then we have the Vehicles Array, the MonoBehaviour will pick the vehicles from this array randomly, you can add or remove Vehicles by modifying this array.



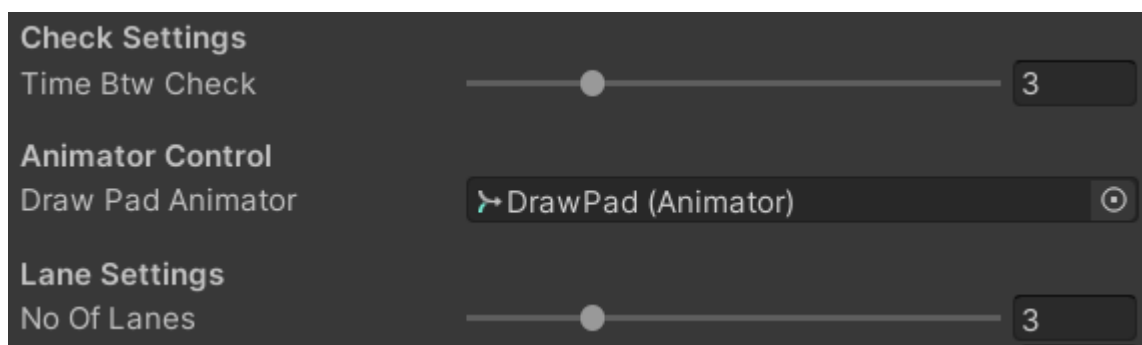
If you want additional Vehicle Variants, you can find them here:

`Assets\Bridge Siege - Template\Prefabs\Cars`

Assets > Bridge Siege - Template > Prefabs > Cars

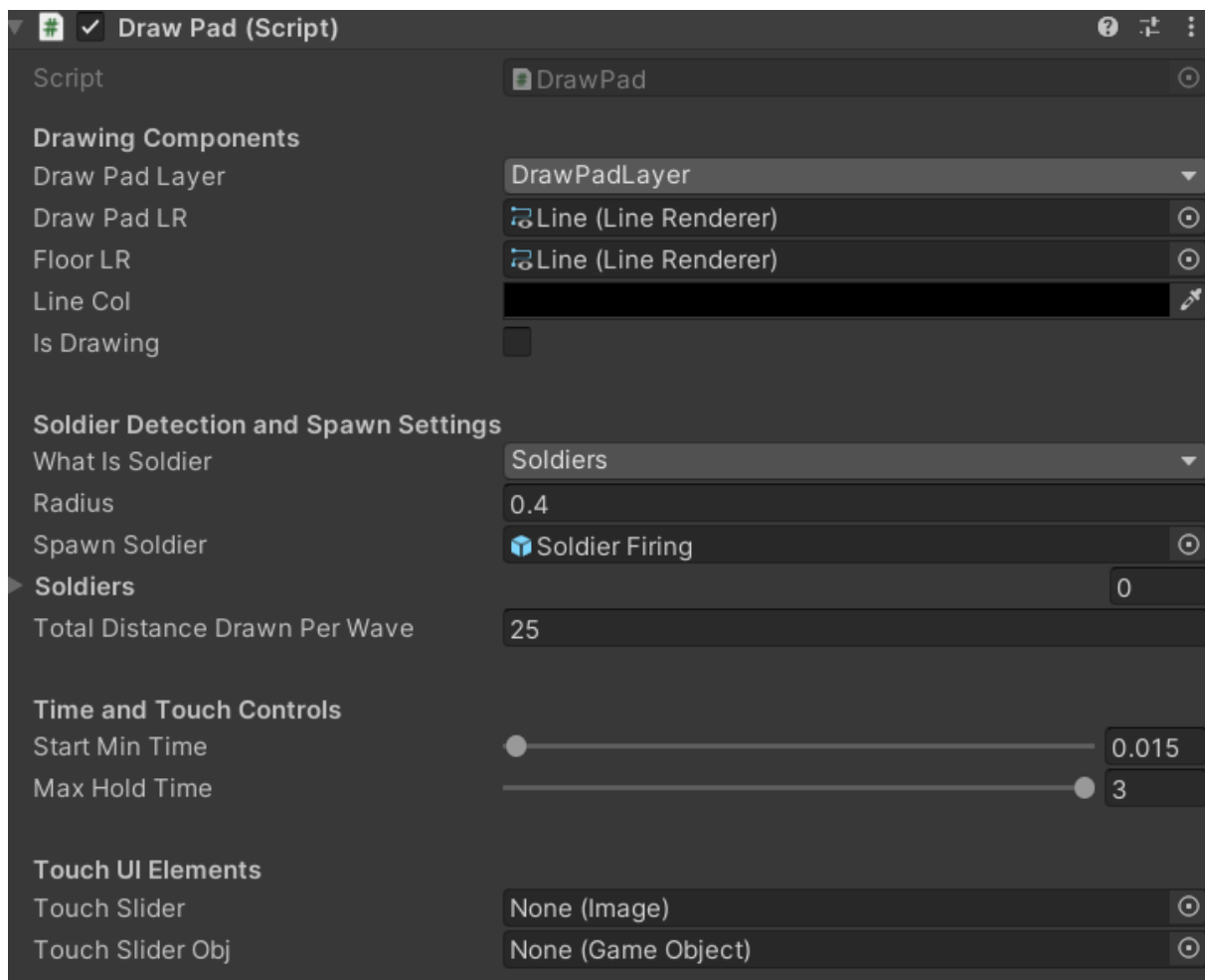


Then we have `No Of Waves` Variable, it shows how many Waves we're gonna have for this level

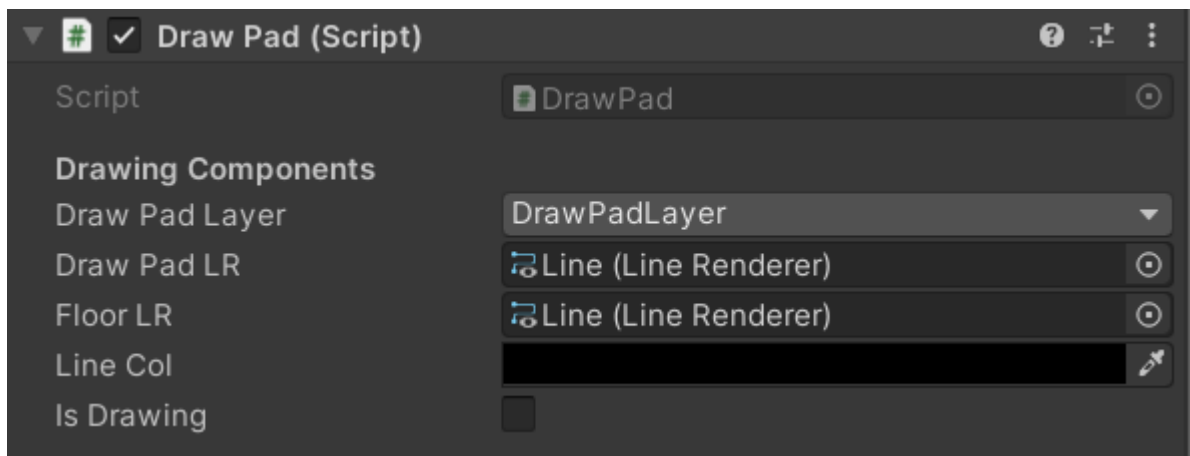


Then we have **Time Btw Check** variable, **Draw Pad Animator** reference to trigger an animation on the Draw Pad, and Lastly a **No Of Lanes** Variable which we have to set according to each level.

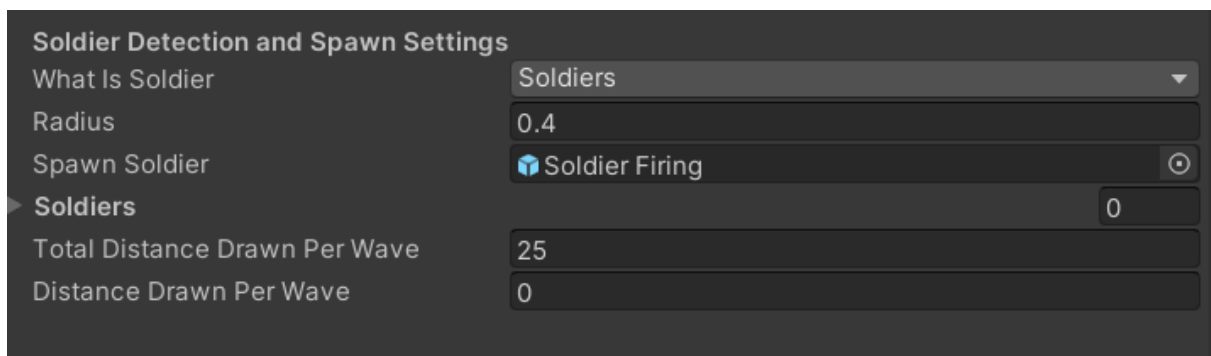
Draw Pad



We have a Draw Pad Floor and Draw Pad UI Project, we are drawing on the UI and then mimicking the same line positions in floor Draw Pad and Spawning the Troops there.



So we are checking whether we are drawing on draw pad or not using Layers, then we have a `Draw Pad LR` (Line Renderer) Reference and a `Floor LR` (Line Renderer) Reference, also you can change the line color using this `Line Col` Setting. Lastly we have a bool variable to check whether we are drawing right now or not.



Now, since we have figured out how to spawn soldiers, there is a problem we can have, that Soldiers can be spawned over other Soldiers. So to fix this we have a Physics based Sphere Overlap check to avoid spawning soldiers over other soldiers.

We again use a Soldier Layer to check for Soldiers and a radius to check for soldiers, then we have a `Spawn Soldier` Prefab reference to which soldier to spawn, we sometimes change this variable, for example in case of rewarded ad we replace this default soldier prefab reference with what a player watched an ad for soldier. Lastly we have Total Distance we can draw per wave to make it challenging for the player by restricting the draw distance.

Time and Touch Controls	
Start Min Time	0.015
Min Time	0.5
Max Hold Time	<input type="range" value="3"/>
Time Holding Mouse	0
Touch UI Elements	
Touch Slider	None (Image)
Touch Slider Obj	None (Game Object)
Reference to GameManager	
Game Manager	None (Game Manager)

Now we also have a time restriction and we control Time Restriction using these two float variables - `StartMinTime` (when player can start drawing again) & `MaxHoldTime` (max drawing time on drawpad)

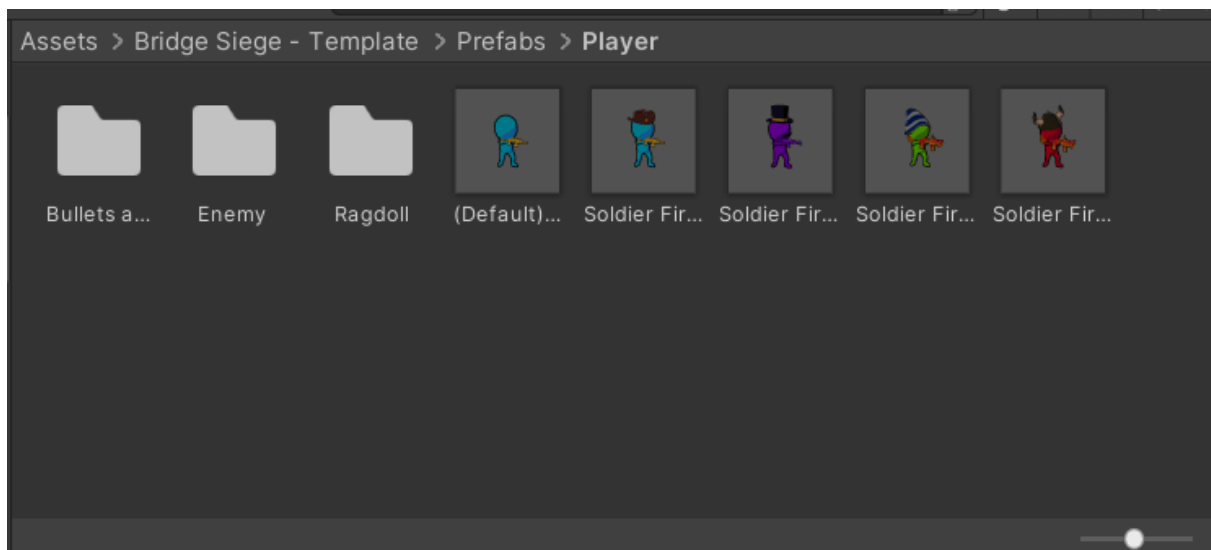
`Touch Slider (image)` - we let the player know how much distance they have drawn till now using this slider UI, `Touch Slider Obj` is just `SetActive (true / false)` through this script.

Soldier

It's responsible for Soldier Death after vehicle collision & Bullet (or Rocket) Firing.

Assets > Bridge Siege - Template > Prefabs > Player

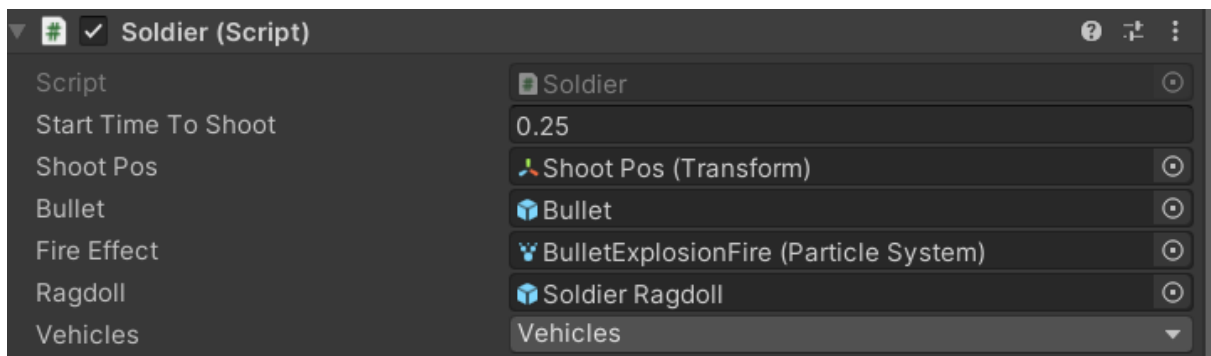
Go to above location and open any `Soldier` Prefab



I am going to open the Default Soldier Prefab



The first child object (Soldier Firing) contains the Soldier Script.



Start Time to Shoot - Its Fire Rate

Shoot Pos - Bullet Spawn Position

Bullet - Reference to Bullet Prefab / Rocket Prefab (in case of Rocket Launcher)

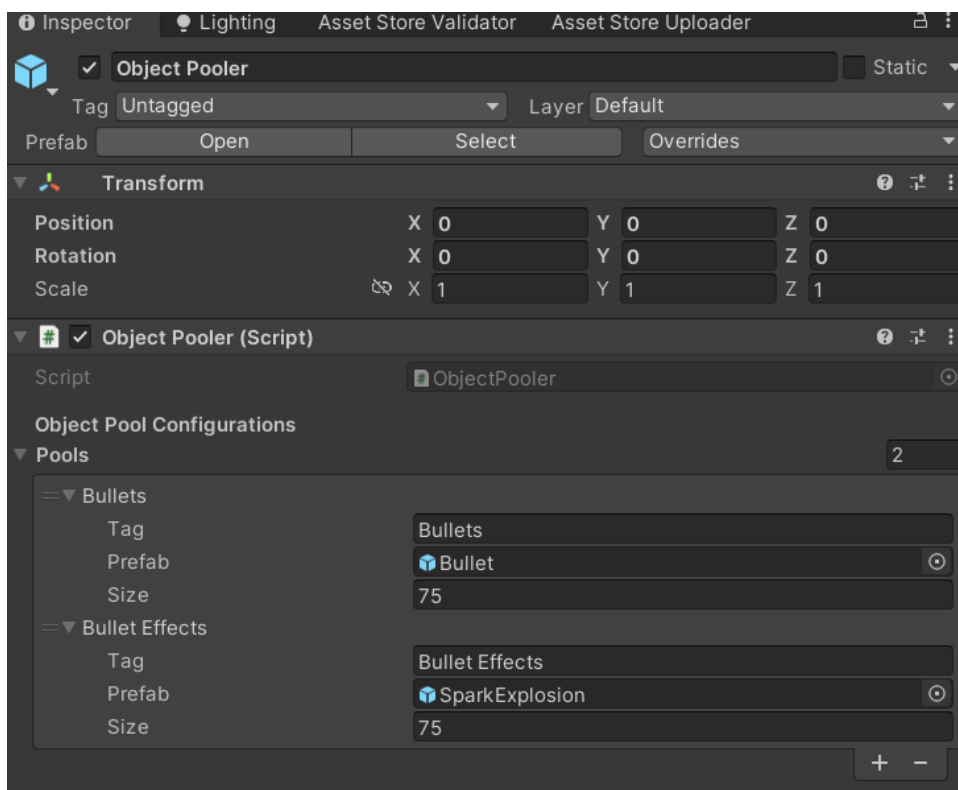
Fire Effect - Muzzle Flash

Ragdoll - Ragdoll Prefab (spawn when player dies)

Vehicles - what is Vehicle detection (through layer)

Object Pooler

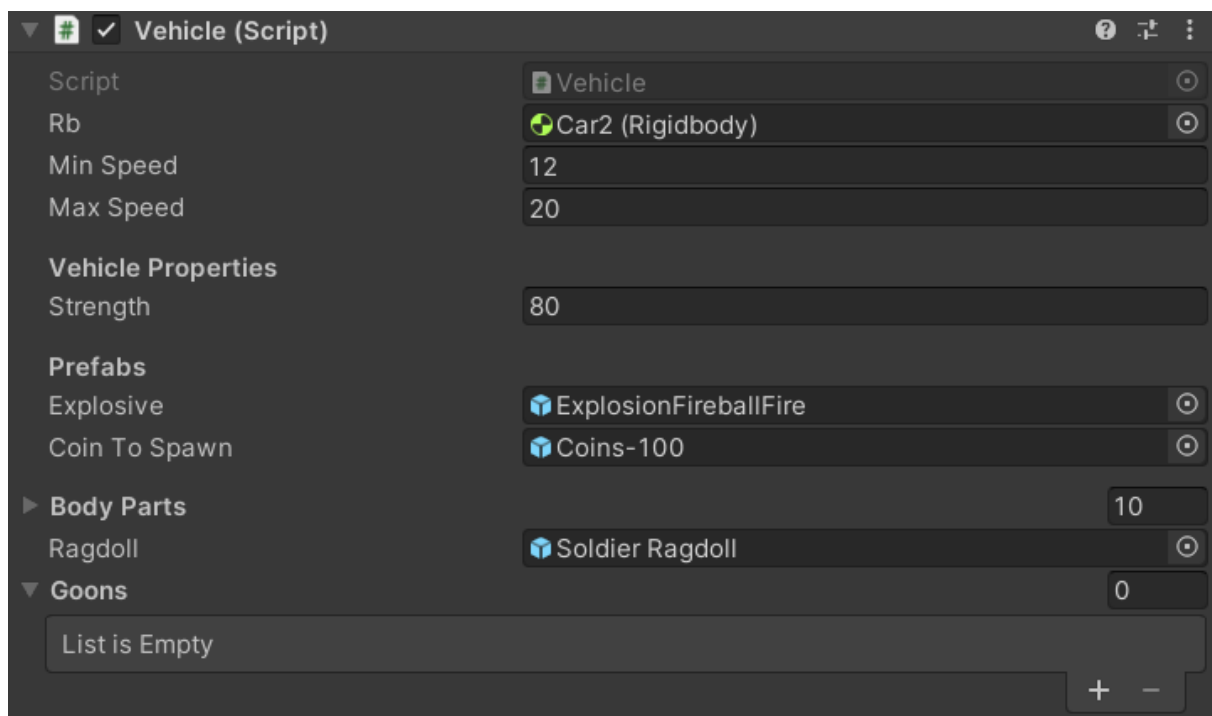
We are using the Object Pooler to optimize the performance of the game, It's Responsible for the spawning of Bullet (or Rockets) and Bullet Effect on collision (or Rocket Explode Effect on collision)



Vehicle

It's attached to every vehicle what soldiers have to face, This Component controls the vehicle forward movement, vehicle explosion, and coin spawning on explosion, if there are some goons in the vehicle who are firing at the

soldiers, then their death upon vehicle explosion is also controlled by this component



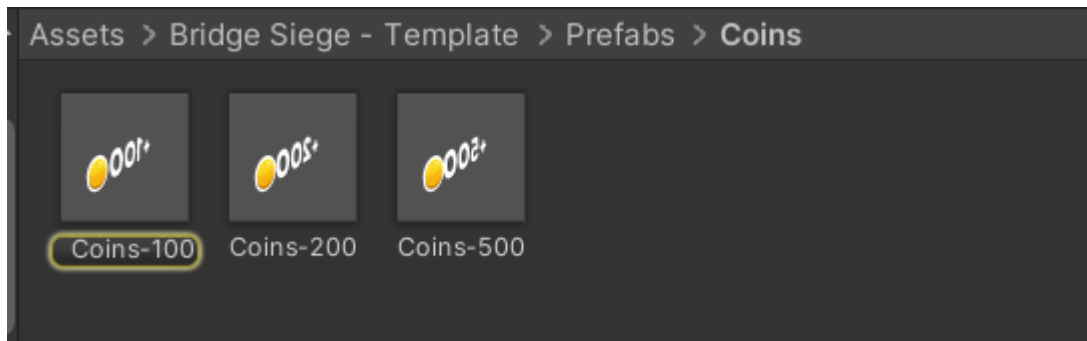
Rb - Reference to the Rigidbody Component of the Vehicle.

Min Speed & Max Speed - The Vehicle will be assigned with a random speed based on these inputs

Strength - Vehicle Strength as in how many bullet hits it can absorb (Strength of a Truck is 150 and a Car's Strength is about 80)

Explosive - The GameObject that will be spawned once the Vehicle Explodes (Strength goes down to Zero)

Coin To Spawn - The Coins amount that will be given to player upon vehicle explosion. You can choose between 100 Coins, 200 Coins and 500 Coins.



Body Parts - Reference for all the Vehicle Parts, they will be assigned with Rigidbody Component and Mesh Collider upon explosion.

Ragdoll - Ragdoll that will be spawned in place of Goons upon vehicle explosion

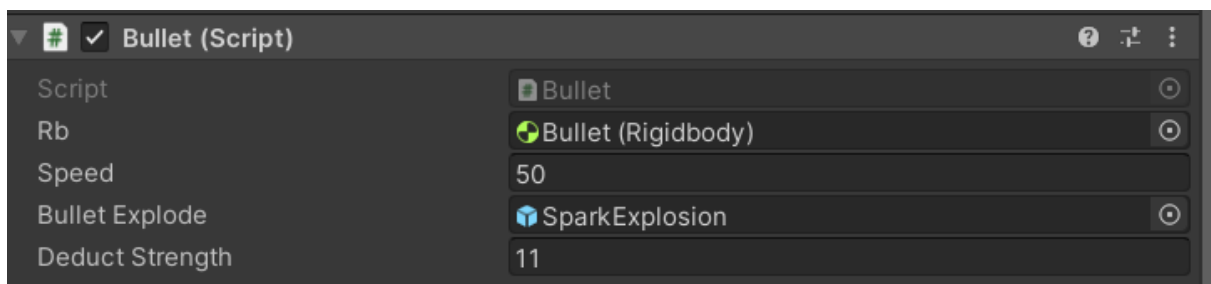
Goons - If you have goons sitting inside the Vehicle put them in this array

Tip : Vehicles that have the Goons are named as "VehicleName" + "ENEMY" inside the folder -

Assets > Bridge Siege - Template > Prefabs > Cars

Bullet

It is attached to the Bullets Object and is responsible for dealing damage to vehicles, spawn collision-hit effects.



Everything is self-explanatory, The **Deduct Strength** is the amount by which vehicle strength will be reduced upon collision.